Reg. No:

SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : Jaasiel. J S

Programme: **B.E / B.Tech** Slot :D

Course Code: **CSA1519** Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics** Date:

Duration **1:30 Hr** Max Marks **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:	
CO1	Analyze cloud computing architectures, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud

Question No	Overall Mark Distribution	Course Outcome	Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Energy Saving	CO1	BL3	10	
3	Security	CO2	BL4	10	
4	and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
	Total			50	

Student Signature	Faculty Signature	Program Director

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25 %
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65 %
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service

sharing between the universities. Examine the federation challenges and the privacy risks associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	**Value**
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25%of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

CIA-1 test

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10)

(i) Development of computing hardware and internet software.

* powerful processors and increased memory improved computing performance.
* High-speed internet enabled remote access to cloud services.
* virtualization enabled multiple virtual machines on a single physical server.
* web services and APIs enabled communication between applications.

(ii) Cloud service models.

	Requirement	service model	Reason.
*	Existing application using virtual machines	Iaas	provides virtual machines with OS and software cont
*	Develop a new application without managing servers	Paas	provides development tools and runtime environments.

Ready made CRM through browser	Saas	provides software directly over the internet.

3) XaaS (Anything as a Service)

soln: Xaas provides additional services through the cloud such as:

- Storage as a Service (STaaS)
- Database as a Service (DBaaS)
- Network as a Service (NaaS)
- security as a Service (SECaaS).

∴ cloud computing helps streamline reduce infrastructure costs, improve scalability and flexibility and access applications from anywhere.

20)

ans: Given:

- No. of physical servers = 15
- cpu utilisation = 25%.
- power consumption per server = 350W
- New client power consumption = 450W

* Target utilisation = 65%.
* Electricity cost = ₹9/KWH.

(a) Number of host required:

Total cpu load:

$15 \times 25 = \boxed{375\%}$.

Actual host required:

$= \frac{375}{65}$.

$= \boxed{6.77}$

Including N+1:

$6+1 \Rightarrow \boxed{7 \text{ Hosts}}$

(b) Consolidation ratio:

$\Rightarrow \frac{15}{6}$

$\Rightarrow \boxed{2.5:1}$

Cpu utilization:

When all 7 host are active:

sufficient

$\Rightarrow \frac{375}{7\times100} \times 100$

$\Rightarrow$ 53.57%.

After one rest fails (6 rests):

$\Rightarrow \frac{375}{6\times100}$

$\Rightarrow$ 62.5%.

d) Annual electricity cost saving:

Existing power consumption:

$15 \times 350 \Rightarrow$ 5250W.

$\therefore \Rightarrow$ 5.25KW

New power consumption:

$7 \times 450 \Rightarrow$ 3150W.

3.15KW

Annual energy saving:

$(5.25 - 3.15) \times 24 \times 365$

$= 18396$ KWh.

Annual rest

18396×9

$\Rightarrow$ ₹ 1,65,564.

3a) Given :

* physical CPU cores = 24
* Total RAM = 96GB
* Hypervisor RAM = 8GB
* Available RAM = 88GB
* Number of VMs = 12
* Each VM = 8 vCPUs and 8GB RAM

(a) Over commitment ratios :

CPU over commitment :

$$\Rightarrow \frac{12 \times 3}{24}$$

$$\Rightarrow \boxed{1.5:1}$$

Memory over commitment :

$$\frac{12 \times 8}{88} \Rightarrow \boxed{1.09:1}$$

(b) Average and peak memory sufficiency

Normal operation :

$$\Rightarrow 12 \times 5$$
$$\Rightarrow 60\,GB$$

Available RAM $\Rightarrow$ 88GB.

$\therefore$ Memory is sufficient

Peak operation:

(8×8) + (4×6)

⇒ 64 + 24 = 88GB

∴ The peak memory is exactly sufficient, with no free memory available for additional workload.

9) Average and peak cpu sufficiency

Normal operation:

12×3 × 35% = 12.6

CPU usage = 12.6 cores.

Available cores = 24

∴ cpu is sufficient.

Peak operation:

⇒ 12×3 × 75%

⇒ 27

CPU demand = 27 cores

Available core = 24

27 - 24

⇒ 3 cores

∴ The cpu is insufficient during peak operation.

d) Performance and security issues with Remedies

Issues	Remedies
* CPU contention	* Add more CPU cores
* Memory exhaustion	* Increase RAM and monitor usage.
* VM performance degradation	* Use resource allocation and load balancing.
* Unauthorized VM access	* Implement strong authentication.
* VM isolation failure.	* Apply hypervisor security patches.

4a)

1) Conventional cloud data centre

* Provides servers, storage and networking resources.
* Virtualization allows resource sharing between universities.
* Support research applications and digital libraries.

2) Software - Defined Data centre (SDDC)

* Manages computing, storage and networking through software.
* Enables automatic resource utilization and allocation
* Improves scalability and workload balancing.

3) Role of CDA

* Connects services using standard APIs
* Enables communication between two universities
* Supports sharing of research and library services.

4) Role of Federation challanges and privacy risks

* Identity theft and unauthorized access.
* User presence information leakage.
* Data privacy and security issues
* Different cloud platforms may have compatibility problems.

5) solutions:

* use multi factor authentication
* Implement Role-Based Access control (RBAC)
* Encrypt data during storage and transmission and data
* Follow data sharing policies and conduct security audits

∴ The cloud federations, SDDC and SOA enable secure resource sharing, better scalability and efficient communication between universities.

5Q) Ans:

Given:

* Cameras: 150
* clip size: 4MB
* Interval = 5 minutes
* Edge filtering = 25%
* Storage cost = ₹1.50 per GB Month.

∴ 540000.

a) Without edge Filtering:

Clips per camera per day:

$$\Rightarrow \frac{24 \times 60}{3}$$

$$\Rightarrow \boxed{480}$$

Total daily clips:

$$150 \times 480 = 72000$$

Daily data volume:

$$72000 \times 4 = 288000 \text{ MB}$$

$$= \boxed{288 \text{ GB}}$$

Average bandwidth:

$$\Rightarrow \frac{288 \times 8}{86400}$$

$$\Rightarrow \boxed{0.02667 \text{ Gbps}}$$

b) 30 Day storage cost

$$288 \times 30 \Rightarrow \boxed{8640 \text{ GB}}$$

⇒ 8640 × 1.50

⇒ ₹12,960

c) With Edge Filtering (25%)

Daily data volume:

⇒ 288 × 0.025

⇒ 7.2 GB

Average bandwidth:

⇒ 26.67 × 0.25

⇒ 6.67 Mbps

Monthly storage cost:

72 × 30 × 1.50 = ₹3,240

Monthly saving:

⇒ 12960 − 3240

⇒ ₹9,720

d) Serverless invocation cost

Monthly invocations:

⇒ 72000 × 30 × 0.25

⇒ 540000.

Cost:

⇒ 840000 × 0.00015

⇒ 281

REST API and Object Storage:

- Rest API: Transfers video clips between cameras and cloud storage using HTTP requests.
- Object storage: Stores huge video files as objects and provides scalability, durability and easy access.